CAPSTONE ANALYTICS PROJECT

Retail Banking Analytics

Turn raw banking data into decisions that protect deposits, reduce risk, and keep customers loyal.

PostgreSQL | Power BI | Microsoft Excel

DOMAIN
Retail Banking

LEVEL
Intermediate – Advanced

FORMAT
End-to-End Project

Student Project Brief | Data Analytics & Business Intelligence Track
You are the analyst. This brief tells you the mission, the rules, and the finish line — the analysis is yours to build.

THE BRIEF

Your Assignment

Scenario — Welcome to Meridian Retail Bank

It's Monday morning. You've just joined the analytics team at **Meridian Retail Bank**, a mid-sized bank with hundreds of thousands of customers, a growing digital app, and a leadership team that is nervous about three things: *Are our best customers about to leave? Is our loan book riskier than we think? And is our money too concentrated in too few hands?*

The Chief Operating Officer drops a set of raw data exports on your desk — customers, accounts, transactions, loans, digital logins, and service complaints — and says: **"I don't want a data dump. I want to know what's happening and what we should do about it. You have the tools. Show me."**

Your job is to become the bridge between messy data and confident business decisions. By the end of this project you will have cleaned the data, interrogated it with SQL, and built dashboards that a busy executive can read in ninety seconds.

> *This is not a "follow the steps and copy the answer" exercise. You are given the questions the business cares about — the queries, logic, visuals, and story are yours to design.*

What the business needs from you

- **Find the money.** Identify high-value customer segments and understand how concentrated the deposit base is.
- **Watch for silence.** Surface dormant accounts and weak engagement before those customers walk out the door.
- **Measure the risk.** Analyse the loan portfolio and expose what actually drives defaults.
- **Follow the clicks.** Track how customers are shifting to digital and mobile channels over time.
- **Judge the service.** Evaluate complaint resolution speed and its link to customer satisfaction.
- **Predict the leavers.** Combine behavioural signals to flag customers at risk of churning.

WHY IT MATTERS

What You'll Walk Away With

This project is deliberately built to look like real work, not a classroom exercise. By the end you'll be able to prove — with artefacts, not just claims — that you can take a business problem end to end.

6
Analytics domains covered, from deposits to churn

3
Executive dashboards designed and built by you

1
End-to-end story: raw data → insight → decision

Skills you'll demonstrate

Technical	Analytical	Business
Advanced SQL (CTEs, window functions, joins, aggregations)	Customer segmentation & cohort thinking	KPI definition & storytelling
Data cleaning & validation in Excel	Risk analysis & ratio building	Translating numbers into recommendations
Dashboard design in Power BI	Trend & growth analysis over time	Designing for a non-technical audience

Portfolio value: A finished version of this project — clean SQL scripts, a short written insight summary, and three polished dashboards — is genuinely strong to show in interviews. Treat it like something you'd put your name on publicly.

GETTING SET UP

Your Toolkit & Your Data

The technology stack

Tool	You'll use it for	What to have ready
Microsoft Excel	First-pass data cleaning, validation, and sanity checks before anything touches the database.	Any recent version. Comfort with filters, pivot tables, and basic formulas.
PostgreSQL	The heart of the project — all business logic, segmentation, and risk analysis lives here.	PostgreSQL installed locally (or a cloud instance) plus a client such as pgAdmin or DBeaver.
Power BI	Turning your query results into interactive dashboards for decision-makers.	Power BI Desktop (free). Connect it to your PostgreSQL database or to exported query results.

Understanding your data model

You'll be working with a set of related tables that mirror how a real bank stores its data. Before writing a single query, spend time understanding how these tables connect — the keys that join them are where all the interesting analysis lives.

Table	What it holds	Key fields to look for
`customers`	One row per customer: demographics, income, join date, location, segment flags.	customer_id, income, join_date, region
`accounts`	Deposit & savings accounts, current balances, account status and open dates.	account_id, customer_id, balance, status, open_date
`transactions`	Individual transactions with date, amount, type, and channel used.	transaction_id, account_id, txn_date, amount, channel
`loans`	Loans issued: principal, interest rate, status, and default flag.	loan_id, customer_id, loan_amount, status, is_default
`digital_activity`	Logins and digital sessions by channel (web, mobile) over time.	customer_id, login_date, channel
`service_tickets`	Complaints/requests with open & resolved timestamps and a satisfaction score.	ticket_id, customer_id, opened_at, resolved_at, csat

Reality check: The exact column names in your dataset may differ slightly from the list above. Real data is never perfectly labelled — part of the job is exploring the tables first and mapping what you actually have to what the business is asking for.

THE PLAN

Project Roadmap — Five Phases

Work through these in order. Each phase produces something you'll build on next, so resist the urge to jump straight to dashboards.

Phase 1 · Explore & Clean

EXCEL

Goal: trust your data before you analyse it.

- Open each file and profile it — row counts, date ranges, obvious duplicates, and blank fields.
- Check for the classics: negative balances that shouldn't exist, impossible dates, duplicate IDs, inconsistent categories (e.g. "Mobile" vs "mobile").
- Document every cleaning decision you make. If you drop or fix a row, write down why.

Phase 2 · Load & Model

POSTGRESQL

Goal: get clean data into the database with the right relationships.

- Design your tables with sensible data types and primary/foreign keys.
- Import your cleaned data and verify the counts match what you had in Excel.
- Write a few simple exploratory queries to confirm the joins behave as you expect.

Phase 3 · Analyse (the core of the project)

POSTGRESQL

Goal: answer the six mission briefs on the next pages using SQL.

- Tackle one analytics domain at a time. Build up from simple aggregates to window-function-driven analysis.
- Keep your queries organised and commented — you'll reuse many of them to feed Power BI.
- Save the output of each key analysis; these become the building blocks for your dashboards.

Phase 4 · Visualise

POWER BI

Goal: build three dashboards a busy executive can actually use.

- Connect Power BI to your database or query exports.
- Build the three required dashboards (specified later in this brief).
- Design for clarity: the headline number first, the detail underneath.

Phase 5 · Tell the Story

YOU

Goal: convert your findings into a short set of recommendations.

- Write a one-page executive summary: the five or six things leadership must know.
- For each insight, state the "so what" — the action you'd recommend.
- Be honest about limitations and what you'd investigate with more data.

THE REAL WORK

Analytical Mission Briefs

These six missions are the substance of the project. Each gives you the **business questions** to answer and the **techniques** worth reaching for — but never the answer. How you write the SQL and what you conclude is entirely up to you.

01 Customer & Segmentation Analytics

Leadership wants to know *who* their customers really are and which ones matter most.

QUESTIONS TO ANSWER

- How can customers be grouped into meaningful value tiers (e.g. by balance or income)?
- Who are the high-value customers, and what share of total deposits do they hold?
- How long have customers been with the bank, and does tenure relate to value?

TECHNIQUES TO REACH FOR — grouping & aggregation, ranking customers, and dividing them into tiers. Think about which window function naturally splits a population into equal-sized buckets.

02 Deposit & Balance Analytics

Concentration risk keeps executives awake at night — what if a handful of customers hold most of the money?

QUESTIONS TO ANSWER

- What is the average balance across the bank, and how does it vary by segment or region?
- How concentrated are deposits? What share is held by the top few percent of customers?
- How many accounts are dormant, and how much balance is sitting idle in them?

TECHNIQUES TO REACH FOR — running totals and cumulative shares, percentile bucketing, and defining a clear rule for what counts as "dormant" based on transaction recency.

03 Loan & Risk Analytics

The COO suspects the loan book is riskier than the headline numbers suggest. Prove it or disprove it.

QUESTIONS TO ANSWER

- What is the overall default rate, and how does it differ across customer groups?
- Build an income-to-loan ratio — does a lower ratio really mean higher default risk?
- Which risk segments deserve the most attention from the credit team?

TECHNIQUES TO REACH FOR — building derived ratios, banding customers into risk tiers, and comparing default rates across those tiers. Careful handling of joins between customers and loans is essential here.

04 Digital Banking Analytics

The board has invested heavily in the app. Are customers actually using it, and is that use growing?

QUESTIONS TO ANSWER

- Which channels (branch, web, mobile) are customers using, and how is the mix shifting?
- What is the month-on-month growth in digital transactions?
- Which customer segments are the most — and least — digitally engaged?

TECHNIQUES TO REACH FOR — aggregating activity by month and channel, then comparing each month to the one before it. Which window function lets you look at the *previous* row's value to compute growth?

05 Customer Service Analytics

Complaints are expensive. Slow ones are worse. Quantify the link between speed and satisfaction.

QUESTIONS TO ANSWER

- What is the average resolution time, and how does it vary by complaint type?
- Is faster resolution associated with higher satisfaction scores?
- Where are the service bottlenecks that hurt the customer experience most?

TECHNIQUES TO REACH FOR — calculating durations between two timestamps, then grouping resolution times into bands and comparing average satisfaction across them.

06 Churn-Risk Identification (Capstone Challenge)

The hardest and most valuable mission: bring signals from every other domain together to flag who might leave.

QUESTIONS TO ANSWER

- What behavioural signals hint that a customer is disengaging (falling balances, no logins, dormancy, unresolved complaints)?
- How would you combine these signals into a single churn-risk view?
- Which high-value customers are also high churn-risk — the ones the bank can least afford to lose?

TECHNIQUES TO REACH FOR — layered CTEs that stitch together outputs from earlier missions, scoring logic, and a final ranking. This is where you show you can think like an analyst, not just query like one.

A note on method: There is rarely one "correct" threshold for dormancy, high value, or risk. Choose sensible cut-offs, *state your assumptions clearly*, and be ready to defend them. Good judgement matters as much as good syntax.

BRING IT TO LIFE

Dashboard Requirements

Your SQL answers the questions; your dashboards make people care. Build these three in Power BI. Design each so the most important number is the first thing the eye lands on.

Dashboard 1 — Executive Summary

A single-screen health check of the whole bank for the COO.

- Headline KPIs: total customers, total deposits, total loans, default rate, digital adoption.
- A high-level view of portfolio performance and the key risk indicators.
- Should answer "how is the bank doing?" in under 90 seconds.

Dashboard 2 — Customer & Deposit Analytics

For the retail and marketing teams who need to act on customers.

- Customer segmentation and value tiers.
- Balance distribution and deposit concentration.
- Dormant accounts — how many, and how much idle balance they represent.

Dashboard 3 — Loan & Risk Analytics

For the credit and risk teams making lending decisions.

- Loan portfolio quality and default behaviour.
- Income-based risk segmentation.
- Clear identification of the segments that need intervention.

Design principles to follow: keep colours consistent and meaningful (e.g. red only for risk), avoid clutter, label everything a non-analyst could misread, and always show the "so what," not just the "what."

Optional 4th dashboard: If you have time, build a Digital & Service dashboard combining channel adoption trends with complaint-resolution and satisfaction metrics. It rounds out the story nicely.

THE FINISH LINE

Deliverables & How You'll Be Judged

What to submit

- ☐ A cleaning log documenting the issues you found and how you handled them (Excel or a short note).
- ☐ A well-organised, commented SQL script (or scripts) covering all six missions.
- ☐ Your Power BI file (.pbix) with the three required dashboards.
- ☐ A one-page executive summary of insights and recommendations.
- ☐ A short README explaining how someone could re-run your work.

Evaluation rubric

Criterion	Weight	What "excellent" looks like
Data preparation	15%	Issues found and handled thoughtfully; decisions documented and justified.
SQL quality & correctness	30%	Correct results, clean structure, effective use of CTEs and window functions.
Depth of analysis	20%	All six missions addressed with real insight, not just surface aggregates.
Dashboard design	20%	Clear, accurate, executive-ready visuals that answer the business questions.
Insight & storytelling	15%	Findings translated into concrete, defensible recommendations.

Pro tips for a standout submission

Comment as you go. Every query should start with the business question it answers. Your future self (and your reviewer) will thank you.

Validate everything twice. If a number looks surprising, it's either a great insight or a bug. Check before you present it.

Stretch goals for the ambitious: add a simple churn-risk score column and rank customers by it; build a reactivation target list for dormant high-value customers; or add a parameter to your dashboards so users can change the "dormancy" or "high-value" threshold themselves.

Academic integrity: The techniques are named for you, but the queries, thresholds, visuals, and conclusions must be your own. Two identical submissions tell the reviewer neither of them did the thinking.